AI and AI-Designed Interventions for Promoting Cooperation in Human Groups

Yihang Qin
qyh_2024@sjtu.edu.cn

1 Introduction

Human cooperation is central to solving collective action problems. Social challenges such as climate change, public health, democratic deliberation, and the management of common-pool resources require individuals to balance short-term self-interest against long-term collective welfare. Recent large-scale evidence suggests that humans are often more cooperative than purely self-interested accounts would predict. For example, Andre et al. [1] studied cooperation in an incentivised two-player experiment across 125 countries and found that about two-thirds of participants chose to cooperate. Importantly, people systematically underestimated others' willingness to cooperate, and correcting these beliefs increased cooperation. This suggests that cooperation is shaped not only by material incentives, but also by beliefs, norms, and information about others.

However, promoting cooperation is not the same as promoting fairness. In networked public goods problems, allocation rules can shape both collective welfare and inequality. Sheng et al. [2] show that uniform allocation can facilitate the spread of cooperation in social networks compared with allocation proportional to potential contributions, but may also concentrate resources among highly connected individuals and leave peripheral individuals worse off. This highlights a central difficulty for cooperation research: mechanisms that increase aggregate cooperation may have unequal or normatively undesirable consequences. Therefore, interventions designed to promote cooperation should be evaluated not only by whether they increase group welfare, but also by whether they are perceived as fair, legitimate, and acceptable by the people affected by them.

Artificial intelligence offers new tools for understanding and intervening in cooperation. In evolutionary game theory, classical strategies such as tit-for-tat, generous tit-for-tat, win-stay-lose-shift, and zero-determinant strategies have provided important insights into repeated interaction, but they cover only a small part of the possible strategy space. Recent work has used multi-agent reinforcement learning to search larger spaces of finite-memory strategies and discover strategies that can outperform existing benchmarks while promoting cooperation and social welfare across different games and population structures [3]. Beyond strategy discovery, AI can also intervene directly in human groups. Tessler et al. [4] showed that an AI mediator can help people find common ground in democratic deliberation. McKee et al. [5] used deep reinforcement learning and graph neural networks to train a social planner that shaped the network structure of human groups playing a cooperation game. Koster et al. [6, 7] extended this approach to a common-pool resource problem, using behavioural models of human players and deep reinforcement learning to design a social planner that promoted sustainable contributions from human participants. More broadly, work on deep mechanism design suggests that modern AI methods can be used to learn social and economic policies for human benefit, while raising important questions about alignment, human preferences, safety, and ethical deployment [8].

Together, this literature suggests that AI may promote cooperation in at least two ways. First, AI can act as an advisor or mediator, shaping how individuals reason about cooperation, reciprocity, and self-protection. Second, AI can act as a social planner, shaping the incentive structure or redistribution mechanism under which people interact. However, several questions remain unresolved. It is unclear whether LLM advice reliably promotes cooperation in repeated social dilemmas, or whether it sometimes encourages defensive and self-interested behaviour. It is also unclear how humans respond to advice from LLMs, including when they need such advice and then follow or reject such advice. Finally, little is known about whether interpretable AI-derived mechanisms could reveal general principles for redistribution in networked human groups. This proposal therefore asks whether AI intervention can promote human cooperation, and under what conditions it does so without undermining fairness, autonomy, and trust. Experiment 1 examines AI as a conversational advisor in networked repeated social dilemmas. Experiment 2 examines AI as a social planner in networked common-pool resource games, testing whether interpretable AI-derived mechanisms are more acceptable and insightful than black-box mechanisms.

2 Research Questions and Methodological Overview

This proposal asks:

> *Under what conditions can AI interventions promote cooperation in networked human groups without undermining fairness, autonomy, and trust?*

Four related questions guide the research:

1. How does LLM advice affect cooperation in repeated social dilemmas?
2. When do people seek, follow, or reject such advice?
3. Can an AI-designed allocation mechanism improve cooperation and resource sustainability in networked common-pool resource games?
4. Are interpretable AI-derived mechanisms more acceptable and informative than black-box policies?

The project combines computational modelling with online behavioural experiments. Experiment 1 examines AI as a conversational advisor: LLM strategies will first be characterised through repeated-game simulations and then tested with human participants against a no-advice baseline. Experiment 2 examines AI as a social planner: human behavioural data will be used to train simulated agents and a graph neural network-based reinforcement learning planner, whose allocation mechanism will subsequently be evaluated with new human participants. The experiments examine progressively stronger forms of intervention: Experiment 1 tests whether AI can influence cooperation while leaving individual incentives unchanged, whereas Experiment 2 tests whether AI can promote cooperation by redesigning those incentives.

Across the project, I will assess both the behavioural effectiveness and psychological acceptability of AI interventions, and examine whether interventions that improve cooperation and collective outcomes are also perceived as fair, trustworthy, transparent, and legitimate.

3 Experiment 1: LLM Advice and Human Cooperation in Networked Social Dilemmas

3.1 Aim

Experiment 1 will test whether advice from large language models promotes or inhibits human cooperation in repeated social dilemmas. In everyday life, people increasingly use AI systems not as autonomous decision-makers, but as conversational advisors. This raises an important question: when individuals face a conflict between self-interest and collective welfare, does consulting an LLM make them more cooperative, or does it encourage strategic self-protection?

3.2 Design

The experiment will have two parts. In Experiment 1a, I will first characterise the behaviour of LLMs in repeated social dilemmas. LLM agents will play repeated donation games (or Prisoner's Dilemma) on a network. For each decision state, including the player's previous action, neighbours' previous actions, and payoff history, I will query the LLM offline repeatedly to estimate a stochastic advice strategy over cooperation and defection. This will allow me to measure whether the LLM's recommendations are unconditionally cooperative, conditionally cooperative, retaliatory, or self-interested. Furthermore, as a LLM baseline, I will use these strategies to conduct multi-round simulations to assess the final level of cooperation, etc.

In Experiment 1b. human participants will play a networked repeated social dilemma. Participants will independently make decision to cooperate or defect in each round as a No-advice baseline.

In Experiment 1c, human participants will play a networked repeated social dilemma. In each round, participants will first make an initial decision to cooperate or defect. They will then have the opportunity to consult an LLM, which will recommend either cooperation or defection and provide a brief explanation. Participants will then make their final decision. This design allows me to compare initial and final decisions within the same round and to measure whether LLM advice changes human behaviour.

The main behavioural outcomes will be initial cooperation rate, final cooperation rate, advice adoption rate, switching from defection to cooperation, switching from cooperation to defection, and cumulative payoff. Psychological outcomes will include satisfaction with the advice, perceived fairness, trust in the AI, perceived autonomy, and willingness to consult the AI again.

3.3 Predictions

I predict that LLM advice will increase cooperation on average, especially in early rounds and in neighbourhoods where previous cooperation has been relatively high. However, I do not expect the LLM to recommend cooperation unconditionally. Instead, its advice may depend on social history: it may encourage cooperation when reciprocity is possible, but recommend defection or self-protection when a participant has repeatedly been exploited by neighbours.

I also predict that participants will selectively adopt LLM advice. They may be more likely to follow the LLM when its recommendation is consistent with their initial intention, when the explanation appeals to fairness or mutual benefit, or when previous AI advice has led to better outcomes. Conversely, participants may reject advice that appears too self-sacrificial, too selfish, or insufficiently sensitive to the network context. Some studies suggest that, given neutral prompts, LLM strategies may be closer to GRIM (cooperation in the first round; once a player defects, they will always defect thereafter). Humans may be less willing than LLMs to adopt unforgiving strategies, especially when such advice conflicts with social norms of forgiveness, leniency, or giving others a second chance.

This experiment will show whether AI can promote cooperation not by directly controlling incentives, but by shaping how individuals reason about cooperation. It will also identify the conditions under which AI advice improves collective outcomes, and the conditions under which it may instead reinforce defensive or self-interested behaviour.

4 Experiment 2: AI Social Planning in Networked Common-Pool Resource Games

4.1 Aim

Experiment 2 asks whether AI-designed allocation mechanisms can improve cooperation and resource sustainability in networked common-pool resource games. It will also test whether interpretable approximations of learned policies retain their behavioural benefits while increasing human acceptance, and whether these policies reveal how effective allocation depends on individuals' positions in a social network.

(This experiment builds on my previous work on reinforcement learning in network common-pool resource games and extends it from simulation to human behavioural experiments.)

4.2 Design

Participants will be assigned to positions in a network and will repeatedly play a common-pool resource game with their local neighbours. In each round, participants decide how much to contribute to or extract from a local shared resource. A social planner then redistributes resources according to the assigned mechanism. The planner does not control participants' choices, but can shape incentives through redistribution. The AI planner will be trained using a human-in-the-loop pipeline. First, behavioural data will be collected from human participants playing baseline networked common-pool resource games. Second, behavioural cloning will be used to train virtual human agents that approximate human decision-making as a function of local resource states, previous contributions, neighbours' behaviour, and individual payoff history. Third, a graph neural network-based reinforcement learning planner will be trained in this simulated human environment. Finally, the learned mechanism will be tested with new human participants. The AI planner will be compared with several baseline mechanisms: fairness-oriented equal allocation, contribution-based proportional allocation, and interpolation baseline in [6]. I will also test an interpretable rule extracted from the AI policy, allowing me to compare a black-box AI planner with a simpler AI-derived mechanism.

The primary behavioural outcomes will be average cooperation level, accumulated resources, and fairness across players. After the game, participants will rate each mechanism on fairness, transparency, trust, legitimacy, and willingness to use it again. Where possible, participants will also vote between alternative mechanisms.

4.3 Predictions

I predict that AI-designed and AI-derived mechanisms will increase cooperation relative to the no-intervention baseline. However, the interpretable AI-derived rule may be preferred over the black-box AI mechanism, even if the two achieve similar levels of cooperation. This would suggest that human acceptance depends not only on outcome quality, but also on whether the intervention is understandable and normatively acceptable.

A second prediction is that behavioural effectiveness and human satisfaction may diverge. A black-box AI planner may perform better under certain conditions, but based on my previous work, its outputs are less robust and predictable to different conditions and unseen environmental states, so user satisfaction may be lower than that of the interpretable AI-derived rule.

A further prediction is that interpretable AI-derived rules may reveal how redistribution should depend on network position. The planner may not treat all individuals identically: instead, it may assign different weights to equality, contribution, and self-protection depending on an individual's degree. For highly connected individuals, relatively egalitarian redistribution may help stabilise cooperation by supporting disadvantaged neighbours and preventing local resource scarcity from spreading. For individuals with intermediate degree, contribution-based redistribution may provide the strongest incentive for continued cooperation. For low-degree individuals, greater self-retention may be necessary to protect them from resource loss and poverty traps. Such findings would turn the AI planner from a purely predictive tool into a source of interpretable hypotheses about how fair and sustainable redistribution should be organised in networked human groups.

5 Expected Contributions and Fit with the HIP Lab

This project will clarify whether LLM advice causally changes human cooperation, whether people adopt or resist retaliatory AI recommendations, and whether AI social planners trained with simulated human agents generalise to real human groups. It will also test whether the most behaviourally effective mechanism is necessarily the one humans find most fair and acceptable, and whether interpretable AI-derived rules can reveal how resource allocation should depend on network position.

The project closely fits the Human Information Processing Lab's work on human decision-making, human preferences over AI systems, and the use of AI to promote cooperation. My background in strategic games, complex networks, reinforcement learning, and computational modelling would allow me to contribute to this programme while extending it toward the psychological mechanisms that determine when humans trust, adopt, or resist AI interventions.

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